Passivity-Based Control with Physics-Informed Structured Learning for Robot Manipulators

Progress report

José Aarón Barbosa Mercado

CINVESTAV-IPN

Advisors: Dr. Wen Yu Liu & Dr. Fernando Castaños Luna

Outline

1. Problem and architecture
2. Nominal model of the 2R manipulator
3. Slotine–Li and the passivity template (Ortega–Spong)
4. Passivity, feedback equivalence, global stabilization (Byrnes–Isidori–Willems)
5. Nominal passivity-based controller and baseline radius
6. DeLaN and its role; span extension
7. Integral-momentum regression (acceleration-free)
8. Spectral normalization and the Lipschitz certificate
9. Online learning law: passivity derivations and robust modification
10. Super-twisting action and the composite Lyapunov function
11. Simulation results, open points, next steps

Problem statement

Fully actuated manipulator, Euler–Lagrange model

$$M(q)\ddot{q} + C(q,\dot{q})\dot{q} + g(q) + F(\dot{q}) + \tau_d = \tau.$$

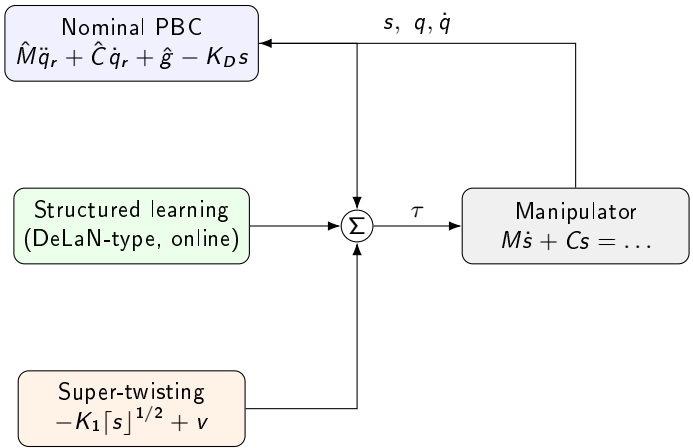
Structural properties (used throughout):

- $\lambda_m I \le M(q) \le \lambda_M I, \quad \dot{M} = C + C^\top$ (Christoffel) $\Rightarrow x^\top(\dot{M} - 2C)x = 0$
- $\dot{q}^\top F(\dot{q}) \ge 0$ (dissipative friction), growth bounds on C, g

Goal: track $q_d(t)$ despite unknown M, C, g, F, τ_d, with:

1. a nominal model giving a *guaranteed* baseline error radius,
2. a structured learning module that shrinks the residual while *preserving* the physics,
3. a robust (super-twisting) action for what remains,
4. a stability argument that is *modular*: each block certified through passivity.

Architecture



Sliding variable $s = \dot{e} + \Lambda e$, $e = q - q_d$; reference $\dot{q}_r = \dot{q}_d - \Lambda e$, $\ddot{q}_r$ its derivative.

The modularity argument

- Adding a weight term to a quadratic storage is *not* unique to passivity (standard adaptive control does it).
- What passivity buys: each block is certified through its own *port* map; the negative feedback of passive systems is passive.
- With the interconnection fixed, **swapping a module only requires re-certifying that module's passivity** (e.g. another learner, another robust term).
- In this thesis the modules are chosen (PBC, DeLaN-type learner, STA), so the immediate benefit is a clean certificate and a physical interpretation of what each block contributes.

Caveat

The STA is *not* passive in the quadratic sense; that block is certified differently.

2R planar manipulator: results

Parameters $m_i,\ l_i,\ l_{c_i},\ I_i;\ c_{12} = \cos(q_1+q_2),\ h = m_2 l_1 l_{c_2} \sin q_2$.

$$M(q) = \begin{pmatrix} m_1 l_{c_1}^2 + m_2(l_1^2 + l_{c_2}^2 + 2 l_1 l_{c_2} \cos q_2) + I_1 + I_2 & m_2(l_{c_2}^2 + l_1 l_{c_2} \cos q_2) + I_2 \\ m_2(l_{c_2}^2 + l_1 l_{c_2} \cos q_2) + I_2 & m_2 l_{c_2}^2 + I_2 \end{pmatrix}$$

$$C(q,\dot q) = \begin{pmatrix} -h\dot q_2 & h\dot q_1 \\ -h\dot q_1 - h\dot q_2 & 0 \end{pmatrix}, \qquad g(q) = \begin{pmatrix} (m_1 l_{c_1} + m_2 l_1) g \cos q_1 + m_2 g l_{c_2} \cos q_{12} \\ m_2 g l_{c_2} \cos q_{12} \end{pmatrix}$$

- Only q_2 enters M; C from Christoffel symbols; $g = \nabla_q P$ with $P = m_1 g l_{c_1} \sin q_1 + m_2 g (l_1 \sin q_1 + l_{c_2} \sin q_{12})$.
- Checked: $\dot M = C + C^\top$ (skew-symmetry of $\dot M - 2C$).
- End effector: $x = l_1 \cos q_1 + l_2 \cos q_{12},\ y = l_1 \sin q_1 + l_2 \sin q_{12}$.

Slotine–Li: the baseline approach

Reference signals: $s = \dot{e} + \Lambda e = \dot{q} - \dot{q}_r, \;\; \dot{q}_r = \dot{q}_d - \Lambda e.$

Linear parametrization: $M\ddot{q} + C\dot{q} + g = Y(q, \dot{q}, \ddot{q})\,\theta.$

Control: $\tau = Y(q, \dot{q}, \dot{q}_r, \ddot{q}_r)\hat{\theta} - K_D s$. With $\tilde{\theta} = \hat{\theta} - \theta$:

$$M\dot{s} + Cs = Y\tilde{\theta} - K_D s.$$

Lyapunov: $V = \frac{1}{2} s^\top M s + \frac{1}{2}\tilde{\theta}^\top \Gamma^{-1}\tilde{\theta},$

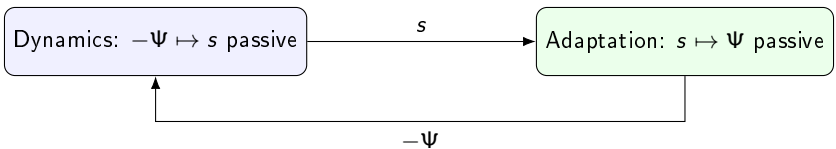
$$\dot{V} = -s^\top K_D s + \tilde{\theta}^\top\big(Y^\top s + \Gamma^{-1}\dot{\hat{\theta}}\big) \overset{\dot{\hat{\theta}} = -\Gamma Y^\top s}{=} -s^\top K_D s \le 0.$$

Barbalat: $s \in \mathcal{L}_2 \cap \mathcal{L}_\infty$, $\dot{s} \in \mathcal{L}_\infty \Rightarrow s \to 0 \Rightarrow e, \dot{e} \to 0$.

Limitation: needs an exact linear-in-parameters structure (no unmodelled dynamics) $\Rightarrow$ motivates the *learned* residual.

Ortega–Spong: passivity as the unifying template

- Error system: $-\Psi$ (identification/regressor error) $\rightarrow$ r is passive; the adaptation law is a passive map $r \rightarrow \Psi$.
- **Theorem 3 (Ortega–Spong '89):** negative feedback of the passive dynamics map and the passive adaptation map gives boundedness and convergence of r.
- Slotine–Li is the special case $K(s) = \Lambda/s$, i.e. $r = F(s)^{-1}e$ with F strictly proper stable.



In this thesis: same template, but $\Psi = \tilde{\Delta}$ (residual) and the passive adaptation block is the output-layer law of a structured network.

Dissipativity and passivity (Byrnes–Isidori–Willems)

$\dot{x} = f(x) + g(x)u, \quad y = h(x)$. Supply rate $w = u^\top y$.

- **Passive:** $\exists\, S \geq 0$: $S(x(t)) \leq S(x(0)) + \int_0^t u^\top y \qquad (\dot{S} \leq u^\top y)$.
- **Lossless:** equality. **OSP:** extra $-\epsilon\|y\|^2$. **ISP:** extra $-\delta\|u\|^2$.
- **Available storage** $S_a(x) = \sup \int_0^T -u^\top y$; finite $\iff$ dissipative; with reachability, S_a is a storage.
- **KYP property:** $L_f S \leq 0, \;\; L_g S = h^\top$. For smooth storage, equivalent to passivity: it is the differential form of $\dot{S} \leq u^\top y$ holding for *all* u.
- **Zero-state detectability + output feedback** $u = -\phi(y)$, $y^\top \phi(y) > 0$: asymptotic stability (LaSalle); global if S is proper.

Relative degree, normal form, minimum phase

- **Relative degree one:** $L_g h(x)$ invertible; $\dot{y} = L_f h + L_g h\, u$ (linear: $\deg(\text{den}) - \deg(\text{num}) = 1$).
- **Normal form:** coordinates (z, y):

$$\dot{z} = q(z, y), \qquad \dot{y} = a(z, y) + b(z, y)u.$$

- **Zero dynamics:** $\dot{z} = q(z, 0)$ (internal dynamics when $y \equiv 0$).
- **Minimum phase:** zero dynamics asymptotically stable (linear: stable inverse). **Weakly minimum phase:** Lyapunov-stable with a C^2 function W, $L_q W \leq 0$.

Manipulator ($y = \dot{q}$)

$\dot{y} = M^{-1}(\tau - \dots)$: relative degree one. Zero dynamics: $\dot{q} \equiv 0$, $q = \text{const}$ (every point an equilibrium) $\Rightarrow$ *weakly* minimum phase. Energy $H = \frac{1}{2}\dot{q}^\top M \dot{q} + U$ gives $\dot{H} = \dot{q}^\top \tau - \dot{q}^\top F \leq \dot{q}^\top \tau$.

Feedback equivalence and stabilization

Theorem (BIW 1991)

(i) A passive system with a C^2 positive definite storage has relative degree one (if $\operatorname{rank} g(0) = m$*) and is weakly minimum phase. (ii) Locally, a system is feedback equivalent ($u = \alpha(x) + \beta(x)v$) to a passive system with C^2 positive definite storage iff it has relative degree one and is weakly minimum phase.*

- Consequence: for such systems, $v = -\phi(y)$ plus zero-state detectability gives asymptotic stabilization; with global versions of the hypotheses and proper storage, **global** stabilization.
- Cascade view (Thm. 5.1): if the y-subsystem is stabilized and the zero dynamics are (weakly) stable, the cascade is stabilized.

Relevance: the manipulator is already passive from τ to $\dot{q}$; the sliding variable $s = \dot{e} + \Lambda e$ is the passive output that adds convergence of e (filtering of a passive output through a stable minimum-phase filter).

Nominal passivity-based controller

$$\tau = \hat{M}\ddot{q}_r + \hat{C}\dot{q}_r + \hat{g} - K_D s \; (+\hat{\Delta}_{\text{learned}} + \tau_{sta}).$$

Define the uncertainty $\Delta = (M - \hat{M})\ddot{q}_r + (C - \hat{C})\dot{q}_r + (g - \hat{g}) + F + \tau_d$ and $\tilde{\Delta} = \Delta - \hat{\Delta}$:

$$M\dot{s} = -Cs - K_D s - \tilde{\Delta}.$$

With $V = \frac{1}{2} s^\top M s$:

$$\dot{V} = -\tfrac{1}{2} s^\top (\dot{M} - 2C)s - s^\top K_D s - s^\top \tilde{\Delta} = -s^\top K_D s - s^\top \tilde{\Delta}.$$

Passivity certificate

$-\tilde{\Delta} \to s$ is **output strictly passive** with storage V and rate $\lambda_{\min}(K_D)$.
Nominal-only: $\hat{\Delta} = 0$, so $\tilde{\Delta} = \Delta$.

Baseline radius

Young with $\Lambda_1 \succ 0$: $-s^\top \Delta \le s^\top \Lambda_1 s + \frac{1}{4}\Delta^\top \Lambda_1^{-1}\Delta$, so

$$\dot V \le -s^\top (K_D - \Lambda_1)s + \bar\eta_1 \le -\alpha V + \bar\eta_1, \quad \alpha = \frac{2\lambda_{\min}(K_D - \Lambda_1)}{\lambda_M}, \quad \bar\eta_1 = \sup_t \tfrac{1}{4}\Delta^\top \Lambda_1^{-1}\Delta.$$

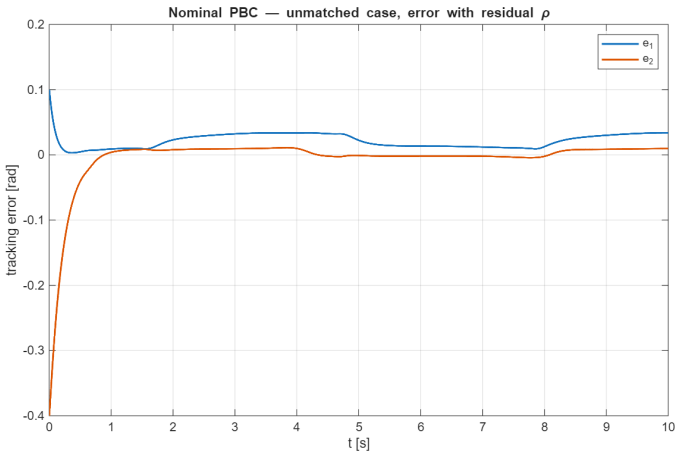
Comparison lemma $\Rightarrow$ ultimate bound

$$\boxed{\limsup \|s\| \le \rho_1 = \sqrt{\frac{\bar\eta_1\, \lambda_M}{\lambda_{\min}(K_D - \Lambda_1)\, \lambda_m}}}$$

and $e, \dot e$ bounded through the stable filter $s = \dot e + \Lambda e$.

- Larger $K_D \Rightarrow$ smaller ρ_1 (confirmed in the sandbox).
- $\bar\eta_1$ is the quantity the learning module is meant to reduce.

Sandbox: nominal PBC with unmatched dynamics



Setup: plant with $m_2 \times 1.2$, $l_{c_2} \times 0.9$ plus joint friction; controller uses the nominal values. $K_D = 20$.

- Error stays in a bounded band (below ~ 0.02 rad).
- Radius grows as K_D decreases.
- Bumps coincide with $\dot{q}_1 = 0$ (Stribeck friction).
- Friction smoothed with $\tanh(\dot{q}/0.01)$.

DeLaN: physics-informed structured learning

Lagrangian $\mathcal{L} = \frac{1}{2}\dot{q}^\top M(q)\dot{q} - U(q)$; Euler–Lagrange gives $M\ddot{q} + C\dot{q} + \nabla U = \tau$. Only M and U need to be learned:

- $\hat{M} = \hat{L}(q;\theta)\hat{L}^\top(q;\theta) + \epsilon I$, $\hat{L}$ lower triangular, non-negative diagonal $\Rightarrow \hat{M} \succeq \epsilon I$.
- $\hat{C}$ from $\hat{M}$ via Christoffel symbols $\Rightarrow \dot{\hat{M}} - 2\hat{C}$ skew by construction.
- $\hat{g} = \nabla_q \hat{U}(q;\psi)$ via automatic differentiation.
- Friction: monotone activations + non-negative weights $\Rightarrow \dot{q}^\top \hat{F} \geq 0$.

Original DeLaN: offline unsupervised/supervised fit of the decomposition.

Role here: supplies *structure-preserving features*. Offline: learn the basis; online: only the output layer adapts (linear in weights) with a Lyapunov law $\Rightarrow$ the learning block remains passive.

Thesis extension: conic combination of frozen bases

Problem: with a single frozen $\bar{L}$, scalar gains reach only multiples of $\bar{L}\bar{L}^\top$ (e.g. $\left(\begin{smallmatrix}4&2\\2&4\end{smallmatrix}\right)$ but never $\left(\begin{smallmatrix}4&1\\1&4\end{smallmatrix}\right)$).
Fix: k frozen factors $\bar{L}_j$ and non-negative weights

$$\hat{M}_\omega(q) = \sum_{j=1}^{k} \omega_j \, \bar{L}_j(q)\bar{L}_j(q)^\top + \epsilon I, \quad \omega_j \geq 0.$$

- PSD and $\geq \epsilon I$ for *every* ω in the cone: structure cannot be destroyed online.
- Any PSD matrix is a conic combination of rank-one projectors $\Rightarrow$ with rich enough $\bar{L}_j$ the span is not a restriction.
- $\hat{C}$ is linear in $\hat{M}$ (Christoffel is linear), hence slaved to the same weights.
- Gravity channel: weights of free sign (potential field).
- Projection onto the convex cone preserves $\dot{V}$ (since $W^* \in \mathcal{W}$ convex): $\mathrm{tr}\{\tilde{W}^\top(\mathrm{Proj}(\cdot) - \cdot)\} \leq 0$.

Residual from limited span is lumped into ε (approximation error).

Acceleration-free regression: integral momentum

Momentum $p = M\dot{q}$ and $\dot{M} = C + C^\top$:

$$\frac{d}{dt}(M\dot{q}) = M\ddot{q} + \dot{M}\dot{q} = \tau - g - F + C^\top \dot{q}.$$

Integrate over a window $[t_1, t_2]$ (no $\ddot{q}$ needed):

$$\boxed{M(q_2)\dot{q}_2 - M(q_1)\dot{q}_1 = \int_{t_1}^{t_2} \left[\tau - g(q) - F(\dot{q}) + C^\top(q, \dot{q})\dot{q}\right] dt}$$

- Regression targets from measured $q, \dot{q}, \tau$ only; integration acts as a low-pass filter.
- Data: Fourier-series excitation trajectories (Swevers et al. 1997) $\rightarrow$ persistent excitation / coverage of the workspace.
- Pipeline implemented in the sandbox: excitation $\rightarrow$ simulated data $\rightarrow$ windowed momentum samples; identity verified with the true parameters.
- Identity from De Luca & Mattone (generalized momenta).

Spectral normalization and the Lipschitz constant

For a feedforward network $f = W^{L+1}\sigma(\cdots\sigma(W^1 x))$ with σ 1-Lipschitz:

$$\|f\|_{\mathrm{Lip}} \le \prod_l \sigma_{\max}(W^l).$$

Normalization (Miyato '18; Neural Lander):

$$\bar{W}^l = \frac{W^l}{\sigma_{\max}(W^l)}\,\gamma^{1/(L+1)} \;\Rightarrow\; \|f\|_{\mathrm{Lip}} \le \gamma.$$

($\sigma_{\max}$ = largest singular value, *not* the activation; power iteration.)

- γ is a design parameter: lower bound from approximation capacity (error ε), upper bound from the stability conditions.
- Neural Lander use: Lipschitz bound on $f(x_1) - f(x_2)$ in the Lyapunov derivative ($\lambda_{\min}(K_v) > L_a\rho$) and, secondarily, training stability.
- **Our use differs:** a bound on $\|\dot{\tilde{\Delta}}\|$ for the STA (next slides).

What the Lipschitz certificate is needed for

Without STA: the weight bound (e-mod) and the $\mathcal{L}_2$/UUB certificate already hold; **no Lipschitz constant needed**.

With STA: need $\|\dot{\tilde{\Delta}}\| \le L_\theta$. Three contributions:

1. plant growth (through M, C, g bounds and $\ddot{q}_r$),
2. network Jacobian: $\dfrac{\partial \Sigma}{\partial x}\dot{x}$, bounded via spectral normalization,
3. adaptation rate $\|\dot{\hat{W}}\|$ (bounded by the law and weight bounds).

Open point

Coriolis and gravity channels are first derivatives of networks, so their *time* derivative needs **second**-derivative bounds (smooth activations). Spectral normalization alone only controls first-order Lipschitz for the inertia-type output channel.

Learning law: exact cancellation

Network output layer: $\hat{\Delta} = \hat{W}\Sigma(x)$, $\Delta = W^*\Sigma + \varepsilon$, $\tilde{W} = \hat{W} - W^*$, so $\tilde{\Delta} = \varepsilon - \tilde{W}\Sigma$.
$V_\ell = \frac{1}{2} s^\top M s + \frac{1}{2}\,\mathrm{tr}\{\tilde{W}^\top K_w^{-1}\tilde{W}\}$:

$$\dot{V}_\ell = -s^\top K_D s - s^\top \varepsilon + \mathrm{tr}\{\tilde{W}^\top (s\Sigma^\top + K_w^{-1}\dot{\hat{W}})\}$$

using $s^\top \tilde{W}\Sigma = \mathrm{tr}\{\tilde{W}^\top s\Sigma^\top\}$ (cyclic invariance; the derivative passes inside the linear trace). Choose

$$\boxed{\dot{\hat{W}} = -K_w\, s\, \Sigma^\top} \quad \text{(projected onto } \mathcal{W}) \;\Rightarrow\; \dot{V}_\ell = -s^\top K_D s - s^\top \varepsilon.$$

- **Which dynamics?** The error dynamics with the *residual*: $\tilde{\Delta} = \varepsilon - \tilde{W}\Sigma$ (the $W^*\Sigma$ part is what is being learned).

Passivity of the learning module and $\mathcal{L}_2$ gain

Input $u = -\varepsilon$, output $y = s$:

$$\dot{V}_\ell = u^\top y - y^\top K_D y \le u^\top y - \rho\|y\|^2, \qquad \rho = \lambda_{\min}(K_D)$$

$\Rightarrow$ **output strictly passive** $(-\varepsilon \to s)$. Young: $u^\top y \le \frac{1}{2\rho}\|u\|^2 + \frac{\rho}{2}\|y\|^2$:

$$\dot{V}_\ell \le \tfrac{1}{2\rho}\|u\|^2 - \tfrac{\rho}{2}\|y\|^2 \;\Rightarrow\; \|s\|_2^2 \le \tfrac{2}{\rho}V_\ell(0) + \tfrac{1}{\rho^2}\|\varepsilon\|_2^2,$$

$$\boxed{\|s\|_2 \le \sqrt{\tfrac{2V_\ell(0)}{\rho}} + \tfrac{1}{\rho}\|\varepsilon\|_2} \qquad (\varepsilon \in \mathcal{L}_2).$$

- Bad learning degrades performance but does not compromise stability.
- Interconnection: the PBC block $(-\tilde{\Delta} \to s)$ and the learner $(-\varepsilon \to s)$ are two views of the *same* port; only the storage/input definition changes, the output s is shared.

Robust modification (e-modification)

$$\dot{\hat{W}} = -K_w\big[s\Sigma^\top + \kappa\|s\|\hat{W}\big], \quad \kappa > 0.$$

Extra term: $-\kappa\|s\|\langle \tilde{W}, \hat{W}\rangle_F$ with $\hat{W} = \tilde{W} + W^*$. Using Cauchy–Schwarz and completing the square,

$$\langle W^*, \hat{W}\rangle_F - \|\hat{W}\|_F^2 \le -\big(\|\hat{W}\|_F - \tfrac{1}{2}\|W^*\|_F\big)^2 + \tfrac{1}{4}\|W^*\|_F^2,$$

so

$$\dot{V}_\ell \le -\lambda_{\min}(K_D)\|s\|^2 + \tfrac{\kappa}{4}\|W^*\|_F^2\|s\| + s^\top(\tau_{sta} - \varepsilon).$$

- Ultimate bound (no STA): $\|s\| \lesssim \sqrt{\frac{\lambda_M}{\lambda_m}}\,\frac{\frac{\kappa}{4}\|W^*\|_F^2 + \bar{\varepsilon}}{\lambda_{\min}(K_D)}$, weights bounded (projection).
- **Passivity status:** bias linear in $\|s\|$, so *not* OSP; it is **semi-passive** (Pogromsky): dissipative outside a ball whose radius is set by $\kappa\|W^*\|_F^2$.
- Trade-off: larger $\kappa \Rightarrow$ stronger drift protection, larger ball.

Super-twisting action and the composite Lyapunov function

$\tau_{sta} = -K_1 \lceil s \rfloor^{1/2} + v, \quad \dot{v} = -K_2 \lceil s \rfloor^0$ (Filippov inclusion at $s = 0$).

With $\eta = v - \tilde{\Delta}$: $M\dot{s} = -Cs - K_D s - K_1 \lceil s \rfloor^{1/2} + \eta, \quad \dot{\eta} = -K_2 \operatorname{sign}(s) - \dot{\tilde{\Delta}}, \ \|\dot{\tilde{\Delta}}\| \le L_\theta.$

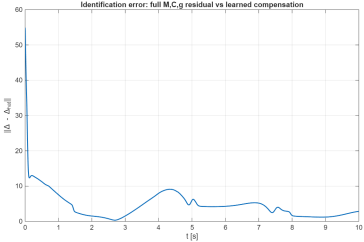
$\zeta = \begin{pmatrix} \lceil s \rfloor^{1/2} \\ \eta \end{pmatrix}, \quad V = \frac{1}{2} s^\top M s + \zeta^\top P \zeta + \frac{1}{2} \operatorname{tr}\{\tilde{W}^\top K_w^{-1} \tilde{W}\}$ (Moreno–Osorio type).

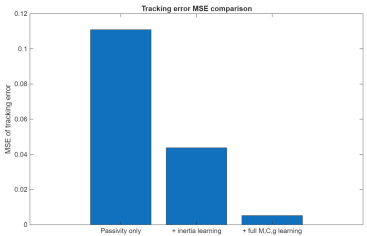
- **Passivity part:** $\frac{1}{2} s^\top M s$ and weight term (plant + learner interconnection).
- **Homogeneity part:** $\zeta^\top P \zeta$; the STA is not passive with a quadratic storage.
- Three terms in $\dot{V}$: plant, network Jacobian (spectral normalization), adaptation rate.

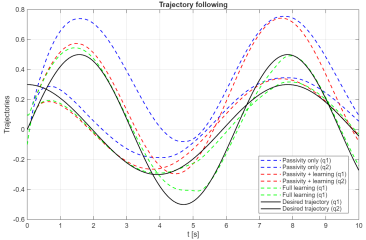
Open: $M^{-1}(q)$ coupling

Route 1: norm bounds with $\kappa_M = \lambda_M / \lambda_m$ and $P = \begin{pmatrix} p_{11} I & p_{12} I \\ p_{12} I & p_{22} I \end{pmatrix}$. Route 2: multivariable STA with robust LMI (Nagesh–Edwards). Deferred until the learning part matures.

Results: PBC only vs. learning







Case with $20 - 40\%$ changes in mass, inertia and center-of-mass location, plus joint friction. The learning module reduces the residual $\tilde{\Delta}$ and the identification error $\|\tilde{\Delta}\|$.
No offline training for the matrices, chosen at random, only online adaptation of the last layer.

Open points and next steps

1. **Offline training** of the DeLaN bases from momentum-window data (loop not yet implemented).
2. **Lipschitz/derivative certificate**: second-derivative bounds for Coriolis and gravity channels; choice of γ.
3. **STA gain conditions** under the $M^{-1}(q)$ coupling (Route 1 vs. Route 2).
4. Simulations: PBC only $\rightarrow$ full learning $\rightarrow$ full learning + STA.
5. Possible changes: IDA-PBC / port-Hamiltonian (regulation only; tracking and matched-uncertainty questions open); reservoir computing.

References I

J.-J. E. Slotine and W. Li, "On the adaptive control of robot manipulators," *Int. J. Robotics Research*, 6(3):49–59, 1987.

R. Ortega and M. W. Spong, "Adaptive motion control of rigid robots: a tutorial," *Automatica*, 25(6):877–888, 1989.

C. I. Byrnes, A. Isidori and J. C. Willems, "Passivity, feedback equivalence, and the global stabilization of minimum phase nonlinear systems," *IEEE TAC*, 36(11):1228–1240, 1991.

R. Ortega, A. van der Schaft, B. Maschke and G. Escobar, "Interconnection and damping assignment passivity-based control of port-controlled Hamiltonian systems," *Automatica*, 38(4):585–596, 2002.

P. Borja and R. Ortega, *Introduction to Passivity-based Control*, 2024.

M. Lutter, K. Listmann and J. Peters, "Deep Lagrangian networks for end-to-end learning of energy-based control for under-actuated systems," *IROS*, 5562–5569, 2019.

G. Shi et al., "Neural Lander: stable drone landing control using learned dynamics," *ICRA*, 2019.

T. Miyato et al., "Spectral normalization for generative adversarial networks," *ICLR*, 2018.

A. De Luca and R. Mattone, "Actuator failure detection and isolation using generalized momenta," *ICRA*, 2003.

References II

J. Swevers et al., "Optimal robot excitation and identification," *IEEE Trans. Robotics and Automation*, 13(5):730–740, 1997.

J. A. Moreno and M. Osorio, "Strict Lyapunov functions for the super-twisting algorithm," *IEEE TAC*, 57(4):1035–1040, 2012.

V. I. Utkin, *Sliding Modes in Control and Optimization*, Springer, 1992.

P. A. Ioannou and J. Sun, *Robust Adaptive Control*, Prentice Hall, 1996.

M. W. Spong, S. Hutchinson and M. Vidyasagar, *Robot Modeling and Control*, Wiley, 2006.

R. Kelly, V. Santibáñez and A. Loría, *Control of Robot Manipulators in Joint Space*, Springer, 2005.

A. Y. Pogromsky, "Passivity based design of synchronizing systems," *Int. J. Bifurcation and Chaos*, 8(2):295–319, 1998.